IMO(4^{th}) – 1962

First Day

1. Find the smallest natural number n with the following properties:
   (a) In decimal representation it ends with 6.
   (b) If we move this digit to the front of the number, we get a number 4 times larger.
2. Find all real number x for which $\sqrt{3-x}-\sqrt{x+1}>\frac{1}{2}$.
3. A cube $ABCDA'B'C'D'$ is given. The point X is moving at a constant speed along the square ABCD in the direction from A to B. The point Y is moving with the same constant speed along the square $BCC'B'$ in the direction from B' to C'. Initially X and Y start out from A and B' respectively. Find the locus of all the midpoints of XY.

Second Day

4. Solve the equation: $\cos^2 x+\cos^2 2x+\cos^2 3x=1$.
5. On the circle k three points A, B and C are given. Construct the fourth point on the circle D such that one can inscribe a circle in ABCD.
6. Let ABC be an isosceles triangle with circumradius r and inradius ρ. Prove that the distance d between the circumcenter and incenter is given by $d=\sqrt{r(r-2\rho)}$.
7. Prove that a tetrahedron SABC has five different spheres that touch all six lines determined by its edges if and only if it is regular.

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